

?? Kontroly shody IEC 61439-1 - Cu 60x5 mm x2

§10.10.3 - Teplotní vzestup, přípojnice ($dTT \leq 105 \text{ K}$)	[OK] 33.7 K (limit 105 K)
§10.10.3 - Teplotní vzestup, svorky ($dTT \leq 70 \text{ K}$)	[OK] 33.7 K (limit 70 K)
Proudová zatížitelnost - $I? \leq I?$ (limit přípojnice)	[OK] 1600 A $\leq$ 2287 A +30.0 %
§10.11 - Zkratová tepelná odolnost ($S \geq S_{\min} = I?^2 \cdot \sqrt{t} / k$)	[OK] 600 mm ² $\geq$ 177.3 mm ² +238.4 %
§10.2 - Nárazová síla - ohybové napětí ($? \leq ??$)	[OK] ? = 126.0 MPa $\leq$ 150 MPa +16.0 %
§10.2 - Pruhyb ($? \leq L?/200$)	[OK] ? = 0.912 mm $\leq$ L/200 = 2.50 mm
§10.10.4 - Úbytek napětí ($\leq 1 \%$ - informační)	[OK] dTU = 0.189 % ($\leq 1 \%$ practice)

[OK] Všechny strukturální kontroly IEC 61439-1 VYHOVUJÍ

?? Klíčové hodnoty

dTT na přípojnici 33.7 K	? provozní 68.7 °C	I? (105 K přípojnice) 2287 A
I? (70 K svorky) 1961 A	S_min zkrat 177.3 mm²	S celkem 600 mm²
I?? 77.78 kA	Síla f 12100.0 N/m	?_max 126.0 MPa
Pruhyb ? 0.912 mm	dTU úbytek napětí 0.754 V (0.189 %)	

Postup výpočtu (IEC 61439-1 / IEC 60865-1)

== INPUTS ==

Material: Cu
Busbar section: 60 x 5 mm -> $A = 300 \text{ mm}^2 \times 2 \text{ bar/phase}$
Cooling perimeter: $P = 2 \times (60 + 5) = 130 \text{ mm}$
Bus length L: 3 m
Installation: flat_v -> $h_{\text{eff}} = 10 \text{ W/(m}^2\text{K)}$
System: AC3 In = 1600 A Un = 400 V
Ambient Ta: 35 °C

== CHECK 1 - TEMPERATURE RISE (IEC 61439-1 §10.10 / Table 5) ==

Steady-state thermal balance per unit length:

$$I^2 \cdot \rho(?) / A = h \cdot P \cdot dTT$$

$R? / \text{m}$ at 35 °C:

$$\begin{aligned} &= \rho?? \cdot (1 + \alpha \cdot (Ta - 20)) / A \\ &= 1.724e-8 \cdot (1 + 0.00393 \cdot (35 - 20)) / 3.000e-4 \\ &= 6.085e-5 \text{ Ohm/m} \end{aligned}$$

$$HC = h \cdot P = 10 \cdot 0.13000 = 1.30000 \text{ W/(m}^2\text{K)}$$

$$I \text{ per bar} = 1600 / 2 = 800.00 \text{ A}$$

$$\begin{aligned} dTT &= I^2 \cdot R? / \text{m} / (HC - I^2 \cdot \alpha \cdot \rho?? / A) \\ &= 800.00^2 \cdot 6.085e-5 / (1.30000 - 800.00^2 \cdot 2.258e-7) \\ &= 33.71 \text{ K} \rightarrow ?_{\text{op}} = 68.7 \text{ °C} \end{aligned}$$

IEC 61439-1 Table 5 limits:

Bare busbar conductors (internal): $dTT \leq 105 \text{ K} \rightarrow [\text{OK}] \text{ PASS}$

Terminals (ext. cable connection): $dTT \leq 70 \text{ K} \rightarrow [\text{OK}] \text{ PASS}$

$$I_z \text{ single bar @ } dTT = 105 \text{ K: } \sqrt{HC \cdot dTT / (R? + dR \cdot dTT)} = 1270.5 \text{ A}$$

$$I_z \text{ single bar @ } dTT = 70 \text{ K (terminals): } = 1089.5 \text{ A}$$

$$\text{Parallel derating } n=2: k_{\text{par}} = 0.9$$

$$I_z \text{ total (2x, busbar limit): } 2286.8 \text{ A}$$

$$I_z \text{ total (2x, terminal limit): } 1961.1 \text{ A}$$

$$In = 1600 \text{ A} \leq I_z = 2286.8 \text{ A} \rightarrow [\text{OK}] \text{ PASS (margin 30.0 \%)}$$

== CHECK 2 - SHORT-CIRCUIT THERMAL WITHSTAND (IEC 61439-1 §10.11) ==

Adiabatic formula (IEC 60865-1): $S_{\text{min}} = I_k \cdot \sqrt{t} / k$

$$k = 141 \text{ A} \cdot \text{s}^{0.5} / \text{mm}^2 \text{ (Cu bare, 20 } \rightarrow 300 \text{ °C per IEC 60364-4-43)}$$

$$S_{\text{min}} = 25000 \cdot \sqrt{t} / 141$$

$$= 25000 \cdot 1.0000 / 141$$

$$= 177.30 \text{ mm}^2$$

$$S_{\text{total}} = 2 \times 300 = 600 \text{ mm}^2$$

$$S_{\text{total}} \geq S_{\text{min}} \rightarrow [\text{OK}] \text{ PASS (margin 238.4 \%)}$$

== CHECK 3 - PEAK FORCE & MECHANICAL WITHSTAND (IEC 61439-1 §10.2) ==

Peak current:

$$? = 2.2 \text{ (IEC 60909 peak factor)}$$

$$I_{\text{pk}} = ? \cdot \sqrt{2} \cdot I_k = 2.20 \cdot 1.4142 \cdot 25000 = 77782 \text{ A (77.78 kA)}$$

Electromagnetic force (2-conductor model):

$$f = 2 \times 10^{??} \cdot I_{\text{pk}}^2 / d_{\text{cc}}$$

$$= 2 \times 10^{??} \cdot 77782^2 / 0.1000$$

$$= 12100.00 \text{ N/m}$$

Simply-supported beam $L_s = 500 \text{ mm}$:

$$\text{Orientation: flat } \rightarrow \text{ STRONG axis (} Z = t \cdot w^2 / 6 = 5 \cdot 60^2 / 6 = 3000.00 \times 10^{??} \text{ m}^3)$$

$$M_{\text{max}} = f \cdot L_s^2 / 8 = 12100.00 \cdot 0.5000^2 / 8 = 378.1250 \text{ N} \cdot \text{m}$$

$$?_{\text{max}} = M_{\text{max}} / Z = 126.04 \text{ MPa vs } ?_y(\text{Cu}) = 150 \text{ MPa}$$

$$?_{\text{max}} \leq ?_y \rightarrow [\text{OK}] \text{ PASS (margin 16.0 \%)}$$

Deflection (limit $L_s/200 = 2.50$ mm):

$$\begin{aligned} \delta &= 5 \cdot f \cdot L^2 / (384 \cdot E \cdot I) \\ &= 5 \cdot 12100.00 \cdot 0.5000^2 / (384 \cdot 1.200 \cdot 10^{11} \cdot 9.000 \cdot 10^{-8}) \\ &= 0.9118 \text{ mm} \\ \delta &\leq L_s/200 \rightarrow [\text{OK}] \text{ PASS (margin 63.5 \%)} \end{aligned}$$

== CHECK 4 - VOLTAGE DROP (IEC 61439-1 §10.10.4) ==

Informational - no explicit IEC 61439 numeric limit; ≤ 1 % is common practice.

$\rho @ \theta_{\text{mean}} = 51.9 \text{ }^\circ\text{C}: 1.940 \cdot 10^{-8} \text{ Ohm} \cdot \text{m}$

$R = \rho \cdot L / (A \cdot n) = 1.940 \cdot 10^{-8} \cdot 3 / (3.000 \cdot 10^{-4} \cdot 2) = 0.0970 \text{ mOhm}$

$X \approx 0.12 \text{ mOhm/m} \cdot 3 \text{ m} = 0.3600 \text{ mOhm} \quad (50 \text{ Hz estimate})$

$dT_U = \sqrt{3} \cdot I_n \cdot (R \cdot \cos\phi + X \cdot \sin\phi) = 0.7540 \text{ V} \quad [3\text{-phase}]$

$dT_U\% = 0.7540 / 400 \cdot 100 = 0.189 \text{ \%} \rightarrow [\text{OK}] \leq 1 \text{ \%}$